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Multi-user MIMO systems and methods

Abstract

A method and system are provided for scheduling data transmission in a Multiple-Input Multiple-Output (MIMO) system. The MIMO system may comprise at least one MIMO transmitter and at least one MIMO receiver. Feedback from one or more receivers may be used by a transmitter to improve quality, capacity, and scheduling in MIMO communication systems. The method may include generating or receiving information pertaining to a MIMO channel metric and information pertaining to a Channel Quality Indicator (CQI) in respect of a transmitted signal; and sending a next transmission to a receiver using a MIMO mode selected in accordance with the information pertaining to the MIMO channel metric, and an adaptive coding and modulation selected in accordance with the information pertaining to the CQI.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7103325	12/2005	Jia	N/A	N/A
7792227	12/2009	Moorti	N/A	N/A
8054898	12/2010	Tong	N/A	N/A
8249518	12/2011	Grant	N/A	N/A
8284852	12/2011	Tong	N/A	N/A
8599945	12/2012	Sampath	N/A	N/A
8611454	12/2012	Tong	N/A	N/A
9301174	12/2015	Tong	N/A	N/A
9313553	12/2015	Sridhar	N/A	N/A
9538408	12/2016	Tong	N/A	N/A
9918328	12/2017	Tong	N/A	N/A

10306658	12/2018	Tong	N/A	N/A
10856305	12/2019	Tong	N/A	N/A
11291017	12/2021	Tong	N/A	N/A
11595089	12/2022	Ramireddy et al.	N/A	N/A
11765763	12/2022	Jia et al.	N/A	N/A
2002/0196752	12/2001	Attar	N/A	N/A
2003/0058881	12/2002	Wu	N/A	N/A
2004/0162098	12/2003	Wei	N/A	N/A
2005/0025254	12/2004	Awad	N/A	N/A
2005/0043031	12/2004	Cho	N/A	N/A
2005/0085195	12/2004	Tong	N/A	N/A
2005/0147177	12/2004	Seo	N/A	N/A
2005/0181739	12/2004	Krasny	N/A	N/A
2005/0201296	12/2004	Vannithamby	N/A	N/A
2006/0013328	12/2005	Zhang	N/A	N/A
2006/0023745	12/2005	Koo	N/A	N/A
2006/0039312	12/2005	Walton	N/A	N/A
2006/0072508	12/2005	Zou	N/A	N/A
2006/0256843	12/2005	Grant	N/A	N/A
2007/0274253	12/2006	Zhang	N/A	N/A
2008/0132281	12/2007	Kim	N/A	N/A
2011/0150117	12/2010	Zhou	N/A	N/A
2013/0028230	12/2012	Borran	N/A	N/A
2016/0100401	12/2015	Xiong	455/450	H04L 25/0242
2017/0170976	12/2016	Ling	N/A	N/A
2020/0220696	12/2019	Nammi	N/A	H04L 1/0026

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2003085875	12/2002	WO	N/A

OTHER PUBLICATIONS

3GPP TS 36.212 V8.8.0; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and Channel Coding; Release 8; Dec. 2009; 60 pages. cited by applicant

3GPP TS 36.213 V8.8.0; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Layer Procedures; Release 8; Sep. 2009; 77 pages. cited by applicant

Office Action dated Jan. 7, 2011; U.S. Appl. No. 12/089,938, filed Apr. 11, 2008; 9 pages. cited by applicant

Notice of Allowance dated Jun. 27, 2011; U.S. Appl. No. 12/089,938, filed Apr. 11, 2008; 7 pages. cited by applicant

Office Action dated Feb. 1, 2012; U.S. Appl. No. 13/251,394, filed Oct. 3, 2011; 10 pages. cited by applicant

Notice of Allowance dated Jun. 11, 2012; U.S. Appl. No. 13/251,394, filed Oct. 3, 2011; 8 pages. cited by applicant

Office Action dated Mar. 5, 2013; U.S. Appl. No. 13/608,234, filed Sep. 10, 2012; 6 pages. cited by applicant

Office Action dated May 30, 2013; U.S. Appl. No. 13/608,234, filed Sep. 10, 2012; 7 pages. cited by applicant

Notice of Allowance dated Sep. 13, 2013; U.S. Appl. No. 13/608,234, filed Sep. 10, 2012; 8 pages. cited by applicant

Office Action dated Aug. 28, 2014; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 7 pages. cited by applicant

Office Action dated Nov. 17, 2014; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 5 pages. cited by applicant

Notice of Allowance dated Feb. 11, 2015; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 7 pages. cited by applicant

Notice of Allowance dated Aug. 14, 2015; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 12 pages. cited by applicant

Office Action dated Feb. 19, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 16 pages. cited by applicant

Final Office Action dated Apr. 29, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 13 pages. cited by applicant

Advisory Action dated Aug. 2, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 2 pages. cited by applicant

Notice of Allowance dated Sep. 15, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 8 pages. cited by applicant

Office Action dated Sep. 15, 2017; U.S. Appl. No. 15/360,600, filed Nov. 23, 2016; 28 pages. cited by applicant

Notice of Allowance dated Dec. 12, 2017; U.S. Appl. No. 15/360,600, filed Nov. 23, 2016; 8 pages. cited by applicant

Office Action dated Sep. 10, 2018; U.S. Appl. No. 15/881,575, filed Jan. 26, 2018; 22 pages. cited by applicant

Notice of Allowance dated Feb. 13, 2019; U.S. Appl. No. 15/881,575, filed Jan. 26, 2018; 8 pages. cited by applicant

Office Action dated Oct. 24, 2019; U.S. Appl. No. 16/384,478, filed Apr. 15, 2019; 21 pages. cited by applicant

Office Action dated Feb. 13, 2020; U.S. Appl. No. 16/384,478, filed Apr. 15, 2019; 8 pages. cited by applicant

Notice of Allowance dated Jul. 22, 2020; U.S. Appl. No. 16/384,478, filed Apr. 15, 2019; 8 pages. cited by applicant

Office Action dated Jul. 2, 2021; U.S. Appl. No. 17/097,907, filed Nov. 13, 2020; 22 pages. cited by applicant

Notice of Allowance dated Nov. 24, 2021; U.S. Appl. No. 17/097,907, filed Nov. 13, 2020; 8 pages. cited by applicant

Office Action dated Jan. 18, 2023; U.S. Appl. No. 17/678,456, filed Feb. 23, 2022; 30 pages. cited by applicant

Notice of Allowance dated May 11, 2023; U.S. Appl. No. 17/678,456, filed Feb. 23, 2022; 8 pages. cited by applicant

PCT International Search Report; Application No. PCT/CA2006/001665; Jan. 24, 2007; 2 pages. cited by applicant

PCT Written Opinion of the International Searching Authority; Application No. PCT/CA2006/001665; Jan. 24, 2007; 6 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/678,456 filed Feb. 23, 2022 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 17/097,907 filed Nov. 13, 2020 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 16/384,478 filed Apr. 15, 2019, now U.S. Pat. No. 10,856,305 issued on Dec. 1, 2020 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 15/881,575 filed on Jan. 26, 2018, now U.S. Pat. No. 10,306,658 issued on May 28, 2019 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 15/360,600 filed on Nov. 23, 2016, now U.S. Pat. No. 9,918,328 issued on Mar. 13, 2018 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 14/832,758 filed on Aug. 21, 2015, now U.S. Pat. No. 9,538,408 issued on Jan. 3, 2017 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 13/971,534 filed on Aug. 20, 2013, now U.S. Pat. No. 9,301,174 issued on Mar. 29, 2016 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 13/608,234 filed on Sep. 10, 2012, now U.S. Pat. No. 8,611,454 issued on Dec. 17, 2013 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 13/251,394 filed on Oct. 3, 2011, now U.S. Pat. No. 8,284,852 issued on Oct. 9, 2012 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 12/089,938 filed on Apr. 11, 2008, now U.S. Pat. No. 8,054,898 issued on Nov. 8, 2011 entitled “Multi-User MIMO Systems and Methods”, which is a filing under 35 U.S.C. § 371 of International Application No. PCT/CA2006/001665 filed Oct. 12, 2006 entitled “Multi-User MIMO Systems and Methods”, claiming priority to U.S. Provisional Application No. 60/725,951 filed Oct. 12, 2005, all of which are incorporated by reference herein as if reproduced in their entirety.

TECHNICAL FIELD

(1) The present application relates generally to communication systems in general, and, more specifically, to MIMO (multiple-input multiple-output) communication systems.

BACKGROUND

(2) In a MIMO communication system, a transmitter transmits data through multiple transmitting antenna ($N_{sub.T}$) and a receiver receives data through multiple receiving antenna ($N_{sub.R}$). The binary data to be transmitted is usually divided between the transmitting antennae. Each receiving antenna receives data from all the transmitting antennae, so if there are M transmitting antennae and N receiving antennae, then the signal will propagate over $M \times N$ channels, each of which has its own channel response.

(3) MIMO wireless communication systems are advantageous in that they enable the capacity of the wireless link between the transmitter and receiver to be improved compared with previous systems in the respect that higher data rates can be obtained. The multipath rich environment enables multiple orthogonal channels to be generated between the transmitter and receiver. Data can then be transmitted over the air in parallel over those channels, simultaneously and using the same bandwidth. Consequently, higher spectral efficiencies are achieved than with non-MIMO systems.

SUMMARY

(4) In some aspects of the present disclosure, a base station in a multi-user MIMO system selects a transmission method on the basis of feedback information received from a plurality of receivers.

(5) In some aspects, the base station assigns a data rate and a MIMO mode suited to the channel quality for that user.

- (6) In some aspects, the present disclosure includes systems and methods which may compute MIMO channel metrics.
- (7) In some aspects, the present disclosure includes systems and methods which may include MIMO mode selection.
- (8) In some aspects, the present disclosure includes systems and methods which may assign/schedule MIMO user transmission and associated formats in order to maximize MIMO communication capacity.
- (9) In some aspects, the present disclosure includes systems and methods which may be used in conjunction with OFDM sub-channels.
- (10) In some aspects, the present disclosure includes systems and methods which use uplink (UL) channel sounding where MIMO matrices may be calculated on the transmission side.
- (11) According to one broad aspect of the present disclosure, there is provided a method comprising: i) generating a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) channel indication, the MIMO channel indication indicating if the MIMO channel is orthogonal; and ii) transmitting a composite metric based on the MIMO channel indication and the CQI.
- (12) According to another broad aspect of the present disclosure, there is provided a transceiver system comprising: i) a generator configured to generate a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) channel indication, the MIMO channel indication indicating if the MIMO channel is orthogonal; and ii) a transmitter configured to transmit a composite metric based on the MIMO channel indication and the CQI.
- (13) According to still another broad aspect of the present disclosure, there is provided a method comprising: i) generating a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) mode indication, the MIMO mode indication indicating a MIMO mode; and ii) transmitting a composite metric based on the MIMO mode indication and the CQI.
- (14) According to yet another broad aspect of the present disclosure, there is provided a transceiver system comprising: i) a generator configured to generate a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) mode indication, the MIMO mode indication indicating a MIMO mode; and ii) a transmitter configured to transmit a composite metric based on the MIMO mode indication and the CQI.
- (15) Other aspects and features of the present disclosure will become apparent to those ordinarily skilled in the art upon review of the following description of the specific embodiments of the present disclosure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will now be described in greater detail with reference to the accompanying diagrams, in which
- (2) FIG. 1 is a block diagram of a multi-user MIMO system in accordance with an embodiment of the present disclosure where channel measurements are calculated on the receive side;
- (3) FIG. 2 is a block diagram of a multi-user MIMO system in accordance with an embodiment of present disclosure where channel measurements are calculated on the transmission side;
- (4) FIG. 3A illustrates a lookup table for an embodiment of the present disclosure in which a Forward Error Correction (FEC) code, a modulation type and a MIMO mode are selected based on a Channel Quality Indicator (CQI);
- (5) FIG. 3B illustrates a key table for FIGS. 3A and 4A;
- (6) FIG. 4A illustrates a lookup table for another embodiment of the present disclosure in which a Forward Error Correction (FEC) code and a modulation type are selected based on a CQI;

- (7) FIG. 4B illustrates a lookup table for a MIMO mode based on a MIMO indicator (MIMO-I);
- (8) FIG. 5 is a diagram illustrating a graphical lookup for a MIMO mode based on a MIMO-I and a CQI;
- (9) FIG. 6 is a plot of BLER (Block Error Rate) versus Signal-to-Noise Ratio (SNR) for an embodiment of the present disclosure;
- (10) FIG. 7 is a plot of BLER (Block Error Rate) versus SNR for another embodiment of the present disclosure;
- (11) FIG. 8A is a plot of Cumulative Distribution Function (CDF) versus condition number of $H_{sup.HH}$ for an embodiment of the present disclosure; and
- (12) FIG. 8B is a plot of CDH versus $H_{sup.HH}$ for another embodiment of the present disclosure.

DETAILED DESCRIPTION

- (13) According to embodiments of the present disclosure, systems and methods are provided which enhance the performance of communication channels in a communication system, to thereby improve, for example, the transmission performance of multi-user MIMO communication systems.
- (14) In multi-user MIMO systems, a multi-data stream transmitter at a base transceiver station (BTS) that provides communication services for a coverage area or cell in a wireless communication system transmits communication signals to a plurality of user terminals via multiple antennas. User terminals are also commonly referred to as MIMO receivers, user equipment (UE), communication devices, and mobile stations, for instance. At a MIMO receiver side, multiple receive antennas are employed for each user.
- (15) FIG. 1 is a block diagram of a multi-user MIMO system in accordance with one embodiment of the present disclosure. On the transmit side, the system of FIG. 1 includes a BTS 100 with an Adaptive Coding and Modulation Module 102, user feedback module 108, and a pair of antennas 110, 112. On the receive side, the system of FIG. 1 includes one or more user terminals 119, 125, 131 (three shown in the illustrated example) include respective MIMO receivers 118, 124, 130 each having a pair of antennas 114, 116, 120, 122, and 126, 128 respectively.
- (16) In user terminal 131, antennas 126, 128 are both connected to the MIMO receiver 130 and to a MIMO channel module 132. MIMO channel module 132 represents the real world radio propagation channel. MIMO receiver 130 is connected to MIMO channel metric measurement module 134. MIMO channel module 132 is connected to CQI metric measurement module 136. Both MIMO channel metric measurement module 134 and CQI metric measurement module 136 are connected to composite feedback module 138. Composite feedback module 138 forms part of the feedback path from the receive side to the transmit side. Information regarding the MIMO channel metric and the CQI metric is transmitted from MIMO channel metric measurement module 134 and CQI metric measurement module 136 respectively to composite feedback module 138 which incorporates one or more lookup tables to determine a MIMO mode and data rate. The MIMO mode and data rate are fed back by composite feedback module 138 to BTS 100 by any convenient communications method, which may or may not comprise wireless communications.
- (17) Each of user terminal 119 and user terminal 125 also include channel measurement modules as well (i.e. each have their own modules equivalent to MIMO channel module 132, MIMO channel metric measurement module 134, CQI metric measurement module 136, and composite feedback module 138). These modules, which are connected to each of MIMO receiver 118 and MIMO receiver 124, have been intentionally omitted to simplify FIG. 1.
- (18) The system of FIG. 1 operates as follows. Pilot data is input into adaptive coding and modulation module 102 where such pilot data is converted into communication signals which may then be transmitted via the antennas 110, 112 from the BTS 100 to user terminals 119, 125 and 131. In some embodiments, the pilots are inserted on each antenna in a manner that makes them distinguishable at a receiver. For example, for OFDM implementations, a respective set of sub-carrier and OFDM symbol durations can be employed for each antenna.
- (19) At MIMO receivers 118, 124, and 130, each of the antennas 114, 116, 120, 126, and 126, 128

receive the pilot signals transmitted from the antennas **110, 112**. MIMO receiver **130** processes the received signals to produce separated layer signals which are fed to MIMO channel metric measurement module **134**. The MIMO channel metric measurement module **134** processes the received pilot data having regard to knowledge of what the transmitted pilot data was, and produces a MIMO channel metric. Specific examples of calculations which may be performed to assess a MIMO channel metric are described below. MIMO channel module **132** processes the received signal to produce MIMO channel state information which is fed to the CQI measurement module **136**. The CQI metric measurement module processes the received pilot data having regard to knowledge of what the transmitted pilot data was, produces a CQI metric. CQI metrics are well known and may for example include CINR (carrier to interference and noise ratio), and the rank of the MIMO channel.

(20) The CQI metric is used as a basis for selecting a particular coding and modulation. BTS **100** can adjust the modulation order and/or coding rate in accordance with the CQI metric. More particularly, the data transmission rate can be increased, decreased, maintained at a constant level, or reduced to 0 bits/s. In a particular example, the CQI is CINR as indicated above, and each range of CINR is associated with a respective adaptive coding and modulation.

(21) In some embodiments, MIMO receivers **118, 124** and **130** track the channel quality via the pilot symbols received and accumulate these quality measurements over a period of time to produce the CQI.

(22) In some embodiments, the feedback from user terminals **119, 125, 131** may also include information identifying the receiver's MIMO capability. For example, this might indicate a number of receive antennas, or the rank of the MIMO channel.

(23) The MIMO channel metric is used to select a MIMO transmission mode to be used for transmitting to a particular user terminal. The particular MIMO modes that are available are selected on an implementation specific basis. Four examples of MIMO modes include beamforming, BLAST, space-time transmit diversity (STTD), and spatial multiplex, though the present disclosure is in no way limited to these MIMO modes and is in fact applicable to all possible space-time mapping.

(24) Those skilled in the art will appreciate that MIMO channel metric measurement and CQI metric measurement may be performed by a digital signal processor (DSP) or a general-purpose processor adapted to execute signal processing software, for example. Various techniques for determining such metric measurements will be apparent to those skilled in the art.

(25) Both the MIMO channel metric and the CQI is transmitted to composite feedback module **138** where one or more lookup tables may be used to determine a composite metric used by BTS **100** to select a MIMO mode and data rate. As used herein, "composite" can be equated to the "overall" quality of the channel matrix. The lookup carried out by composite feedback module **138** is used for two purposes: (i) User terminal pairing, i.e. scheduling. The more orthogonal the channel, the larger the MIMO capacity; and (ii) together with SNR, the lookup is used for MIMO mode and coding and modulation selection. With a higher SNR and composite metric, spatial multiplexing and higher modulation and coding rates may be selected. With a lower SNR and composite metric, transmit and lower modulation and coding rates may be selected.

(26) Note that the composite metric does not affect modulation and coding rates selection in transmit diversity, but it affects modulation and coding rates selection in spatial multiplexing. This is because when the composite metric is low, more inter-layer interference will occur, and hence only lower modulation and coding rates are to be used.

(27) The composite metric is then transmitted by composite feedback module **138** to BTS **100** through user feedback module **108**. With the composite metric received from composite feedback module **138**, a scheduler which forms part of BTS **100** determines a MIMO transmission mode and a modulation and coding to be used for each MIMO receiver. In some embodiments, the BTS **100** indicates the transmission format to each MIMO receiver.

(28) In some embodiments, a two bit composite metric is used, with one bit of the composite metric being used to indicate the CQI, and one bit of the composite metric to indicate the MIMO mode, e.g. transmit diversity or spatial multiplexing. In the spatial multiplexing mode, one additional bit can be used to indicate if the MIMO channel is orthogonal.

(29) FIG. 2 is an alternative embodiment to that illustrated in FIG. 1. In this embodiment, channel metrics are measured at the transmit side **250** rather than the receiver side **201**. On the transmit side **250**, the system of FIG. 2 includes a BTS **200** with an Adaptive Coding and Modulation Module **202**, user feedback module **208**, and a pair of antennas **210,212**. On the receive side **201**, the system of FIG. 2 includes one or more user terminals **219, 225, 231** (three shown in the illustrated example) include respective MIMO receivers **203, 204, 206** each having a pair of antennas **214,216, 220,222, and 226,228** respectively.

(30) In user terminals **219, 225, and 231**, antennas **214,216, 220,222, and 226,228** respectively are connected to MIMO receivers **203, 204, and 206** which each perform UL channel sounding. In the case of Time Division Duplex (TDD), channel sounding is used to allow BTS **200** to perform channel measurements at the transmit side **250** rather than the receiver side **201**. Information received from MIMO receivers **203, 204, and 206** is fed back through a feedback control channel to user feedback module **208** at BTS **200** by any convenient communications method, which may or may not comprise wireless communications.

(31) User feedback module **208** is connected to both MIMO channel metric measurement module **234** and CQI metric measurement module **236**. Both MIMO channel metric measurement module **234** and CQI metric measurement module **236** are connected to composite feedback module **238**. Composite feedback module **238** is connected to adaptive coding and modulation module **202**.

(32) Except for the fact that channel measurements are performed at the transmit side **250** rather than the receive side **201**, the operation of the system of FIG. 2 is otherwise similar to the operation of the system of FIG. 1. The main difference is that through channel sounding, user terminals **219, 225, and 231** pass the burden of channel measurements and processing to BTS **200**.

(33) Of course, the systems of FIGS. 1 and 2 are only two illustrative examples of systems in which the present disclosure may be implemented. The present disclosure is in no way limited thereto. Extension of the principles of the present disclosure to systems having other dimensions will be apparent to those skilled in the art. In particular, the number of user terminals that will be present in a given implementation may differ, and may vary over time if they are mobile. The number of antennas on the base station and user terminals is two in the illustrated example. More generally, any number, two or more, of antennas can be employed such that MIMO communications are possible, though the number of receive antennas must be greater than or equal to the number of data streams (i.e. layers) being transmitted. Both the base station and user terminals include functionality not shown as would be understood to one of skill in the art. Separate components are shown for each of the MIMO channel **132**, MIMO channel metric measurement module **134,234**, the CQI metric measurement module **136,236**, and the composite feedback module **138,238**. More generally, the functions provided by these modules may be combined in one or more functional elements, and these may be implemented in one or a combination of software, hardware, firmware etc.

(34) A MIMO system can be expressed as

$$\mathbf{y} = H\mathbf{x} + \mathbf{n}$$
where $\mathbf{y}$ is a vector of communication signals received at a receiver;

$\mathbf{x}$ is a vector of communication signals transmitted by a transmitter; $\mathbf{n}$ is a vector of noise components affecting the transmitted communication signals;

$$(35) H = \begin{bmatrix} h_{11} & h_{12} & \text{.Math.} & h_{1M} \\ h_{21} & h_{22} & \text{.Math.} & h_{2M} \\ \text{.Math.} & \text{.Math.} & \ddots & \text{.Math.} \\ h_{N1} & h_{N2} & \text{.Math.} & h_{NM} \end{bmatrix} \text{ is a channel matrix of communication channel}$$

attenuation factors;

(36) N is a number of antennas at the receiver; and

(37) M is a number of antennas at the transmitter.

(38) For a [2Tx, 2Rx] MIMO channel,

$$(39) H = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

(40) The eigenvalue of $H \cdot \text{sup.HH}$ are $\lambda_{\max}$, $\lambda_{\min}$. There are several scheduling approaches, including orthogonality and capacity. These approaches are for user terminal pairing only.

(41) Where it is desired that scheduling by a BTS (such as BTS **100** and BTS **200** in FIGS. **1** and **2** respectively) be provided on the basis of maximum orthogonality, the following MIMO channel metric will be computed by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$(42) \max \left\{ \frac{\det(H^H H)}{\text{trace}(H^H H)} \right\}$$

(43) The larger the metric, the more orthogonal is the channel.

(44) For a maximum orthogonality decomposition scheduling scheme,

$$(45) \min \text{.Math.} \begin{bmatrix} h_{11} & h_{21} \end{bmatrix} \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} \text{.Math.} = 0$$

(46) In this case, the channel is completely orthogonal, yielding two separate spatial channels, with channel attenuation factors being $\sqrt{|h_{11}|^2 + |h_{21}|^2}$ and $\sqrt{|h_{12}|^2 + |h_{22}|^2}$ respectively.

(47) For scheduling based on a best conditional number of MIMO channel scheme, the following channel metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\rho = \lambda_{\max} / \lambda_{\min} \approx 1$$

(48) In this case, an advanced receiver (maximum likelihood detection) and/or a simplified receiver can be employed.

(49) For scheduling based on maximum capacity, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\det(H \cdot \text{sup.HH})$$

(50) Maximum capacity scheduling is also maximum CQI scheduling.

(51) For scheduling based on maximum orthogonality for several MIMO channels,

$$(52) \begin{aligned} H_1 &= \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \\ H_2 &= \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \end{aligned} \text{ the following metric will be calculated by, for example, MIMO channel}$$

metric measurement module **134** in FIG. **1**:

$$(53) \max \left\{ \text{.Math.} \frac{\det(H_i^H H_i)}{\text{trace}(H_i^H H_i)} \right\}$$

(54) For scheduling based on orthogonality capacity for several MIMO channels, the following

metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$(55) \max\{ \prod_{i=1}^2 \text{Math. det}(H_i^H H_i) \}$$

(56) For scheduling based on a combined conditional number for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$(57) \min\{ \prod_{i=1}^2 \text{Math. } \kappa_i \}$$

(58) For scheduling based on a SNR weighted maximum orthogonality scheme for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$(59) \max\{ \prod_{i=1}^2 \text{Math. SNR}_i \frac{\text{det}(H_i^H H_i)}{\text{trace}(H_i^H H_i)} \}$$

(60) For scheduling based on a SNR weighted capacity scheme for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$(61) \max\{ \prod_{i=1}^2 \text{Math. SNR}_i \text{det}(H_i^H H_i) \}$$

(62) For scheduling based on a SNR weighted combined conditional number scheme for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$(63) \min\{ \prod_{i=1}^2 \text{Math. SNR}_i \kappa_i \}$$

(64) FIG. **3A** illustrates a table setting out one representative example of a lookup table for selecting coding modulation and MIMO modes based CQI which can be used in accordance with one embodiment of the present disclosure. The table in FIG. **3B** provides a key for the table in FIG. **3A**.

(65) In the table of FIG. **3A**, Row 1 lists possible CQIs, which in this case would be from 1 to 10. Each CQI has an associated FEC code, an associated modulation, and an associated MIMO mode, in this case an STC code. In the particular example, illustrated, there are five available FEC codes, three available modulations, and two available STC codes.

(66) The key shown in FIG. **3B** indicates the code rates $\frac{1}{5}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{4}{5}$ associated with the five available FEC codes, indicates the modulation constellations 4-QAM, 16-QAM 64-QAM associated with the three modulations, and indicates the STC modes STTD and BLAST associated with the two available STC modes.

(67) FIG. **3A** shows how CQI, together with STC code rate, determines the proper code and modulation set. The table in FIG. **3A** is of course just one possible example of a lookup table. The particular MIMO modes, FEC codes, and modulations supported will vary on an implementation specific basis. The present disclosure is in no way limited thereto. Extension of the principles of the present disclosure to others possible lookup tables will be apparent to those skilled in the art. For example, other forms of lookup tables could be employed which include other standard modulation schemes such as Phase-shift keying (PSK), and other forms of MIMO modes such as beamforming.

(68) FIGS. **4A** and **4B** illustrate a lookup table in which the CQI and MIMO are fed back in separate feedback components. In this case, the MIMO mode is determined by both CQI and MIMO.

(69) FIG. **4A** sets out one example for coding modulation based on CQI. Row 1 lists possible CQIs which may be calculated by CQI metric measurement module **136**, from 1 to 10. Row 2 lists Forward Error Correction Codes (FEC), which in this case are from 1 to 5. Row 3 lists three possible forms of modulation which can be implemented in accordance with this lookup table, namely 4-QAM, 16-QAM, and 64-QAM. Reference may be had to the table in FIG. **3B** which

provides a key for the table in FIG. 4A.

(70) FIG. 4B is a lookup table for MIMO modes based on a MIMO indicator (MIMO_I). MIMO indicator may be calculated by MIMO channel metric measurement module 134 shown in FIG. 1. In this case, Row 1 lists possible MIMO indicators from 1 to 10 which are selected based on the result of the channel metric calculations. Row 2 lists possible MIMO formats (which could represent, e.g. BLAST, STTD, beamforming, spatial multiplexing, etc.) from 1 to 5.

(71) FIGS. 4A and 4B show that the higher the CQI/MIMO_I, the higher is the modulation order and FEC coding rates. There are two situations where a composite metric of both CQI and MIMO_I may be used: the first is MIMO mode selection, because when MIMO_I is low, SM may not be used even with a large CQI. FIG. 8 shows that at low MIMO_I, in the high CQI region, the use of STTD or SM as a MIMO mode depends on MIMO_I. However, in the same CQI region, when MIMO_I is larger, SM is selected. Another situation is modulation and code set selection in SM, because in SM, MIMO_I indicates inter-layer interference, and hence indicates the performance of the channel. Given the same CQI, modulation and code set will depend on the MIMO indicator. In other words, modulation and code set will be determined by two parameters: CQI and MIMO_I.

(72) As with the table in FIG. 3A, the tables in FIGS. 4A and 4B are merely illustrative examples of lookup tables which could be used in accordance with the present disclosure. Persons skilled in the art will appreciate that other combinations of parameters can be employed.

(73) FIG. 5 is a diagram illustrating another example of a lookup for MIMO modes based on a MIMO indicator and a CQI. As noted above, a MIMO_I may be ascertained by calculations performed by MIMO channel metric measurement module 134 shown in FIG. 1 and CQI may be calculated by CQI metric measurement module 136 also shown in FIG. 1.

(74) In FIG. 5, three sections are shown, a first section labelled “STTD”, a second section labelled “spatial multiplexing (SM)”, and a third section labelled “STTD/SM”. According to the lookup of FIG. 5, for low CQIs (i.e. low channel quality), regardless of the MIMO indicator, STTD should be chosen as the MIMO format. For high MIMO indicators and high CQIs, the lookup of FIG. 5 indicates that spatial multiplexing should be chosen as the MIMO format. For a low MIMO indicator and high CQI, either of STTD and spatial multiplexing can be selected as the MIMO format.

(75) For the example of FIG. 5, a CQICH (channel quality indicator channel) can be used to provide feedback coding/modulation information and/or selection, and a single bit can be used to flag the MIMO mode. One of ‘0’ or ‘1’ can be used for STTD, and the other of ‘0’ or ‘1’ can be for SM.

(76) FIG. 5 is of course just one possible example of a lookup in which the present disclosure may be implemented. The present disclosure is in no way limited thereto. Extension of the principles of the present disclosure to other lookup diagrams will be apparent to those skilled in the art.

(77) FIG. 6 is a plot of BLER (Block Error Rate) versus SNR (Signal-to-Noise Ratio) for an embodiment of the present disclosure. FIG. 6 shows the impact on maximum capacity scheduling. FIG. 7 is a plot of BLER (Block Error Rate) versus SNR (Signal-to-Noise Ratio) for another embodiment of the present disclosure. FIG. 7 shows the impact on maximum capacity scheduling. Both FIGS. 6 and 7 show two sets of curves, namely, “without scheduling” and “with scheduling”. More particularly, “without scheduling” means “random scheduling”, and “with scheduling” means “orthogonality based scheduling”.

(78) FIG. 8A is a plot of Cumulative Distribution Function (CDF) versus condition number of $H_{\text{sup.HH}}$ for an embodiment of the present disclosure. $H_{\text{sup.HH}}$ means Hermitian transform, i.e. conjugate transposition. FIG. 8A presents the impact on minimum conditional number scheduling. FIG. 8A shows the CDF of the composite metric when random pairing is used (the solid line) and when the orthogonality based pairing is used.

(79) FIG. 8B is a plot of CDH versus (HHH) for another embodiment of the present disclosure. FIG. 8B presents the impact on minimum conditional number scheduling.

(80) Numerous modifications and variations of the present disclosure are possible in light of the

above teachings. It is therefore to be understood that within the scope of the appended claims, the present disclosure may be practised otherwise than as specifically described herein.

Claims

1. A base station comprising: a transceiver; and a processor operationally coupled to the transceiver; wherein the processor is configured to: receive, via the transceiver, channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and transmit, via the transceiver, a MIMO transmission using a number of transmission antennas based on the CSI feedback, wherein the number of transmission antennas is based on the rank and an adaptive coding and modulation type for the transmission selected is based on the CQI.
2. The base station of claim 1, wherein a data transmission rate of the MIMO transmission is based on the selected adaptive coding and modulation type.
3. The base station of claim 2, wherein data transmission rates of subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the received CSI feedback.
4. The base station of claim 1, wherein the MIMO transmission is an Orthogonal Frequency Division Multiplexing (OFDM) transmission.
5. The base station of claim 1, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.
6. A method for use by a base station, the method comprising: receiving channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and transmitting a MIMO transmission using a number of transmission antennas based on the CSI feedback, wherein the number of transmission antennas is based on the rank and an adaptive coding and modulation type for the transmission is selected based on the CQI.
7. The method of claim 6, wherein a data transmission rate of the MIMO transmission is based on the selected adaptive coding and modulation type.
8. The method of claim 7, wherein data transmission rates of the subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the received CSI feedback.
9. The method of claim 6, wherein the MIMO transmission is via an Orthogonal Frequency Division Multiplexing (OFDM) transmission.
10. The method of claim 6, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.
11. A user equipment (UE) comprising: a transceiver; a processor, operatively coupled to the transceiver, wherein the transceiver and the processor are configured to: transmit, via the transceiver, channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and receive, via the transceiver, a MIMO transmission based on the CSI feedback, wherein the rank is used for selecting a number of transmission antennas for the MIMO transmission and the CQI is used for selecting an adaptive coding and modulation type for the MIMO transmission.
12. The UE of claim 11, wherein a data transmission rate of the MIMO transmission is based on the CSI feedback.
13. The UE of claim 12, wherein data transmission rates of the subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the transmitted CSI feedback.
14. The UE of claim 11, wherein the MIMO transmission is an Orthogonal Frequency Division Multiplexing (OFDM) transmission.

15. The UE of claim 11, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.
 16. A method for use by a user equipment (UE), the method comprising: transmitting channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and receiving a MIMO transmission based on the CSI feedback, wherein the rank is used for selecting a number of transmission antennas for the MIMO transmission and the CQI is used for selecting an adaptive coding and modulation type for the MIMO transmission.
 17. The method of claim 16, wherein a data transmission rate of the MIMO transmission is based on the CSI feedback.
 18. The method of claim 17, wherein data transmission rates of the subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the transmitted CSI feedback.
 19. The method of claim 16, wherein the MIMO transmission is an Orthogonal Frequency Division Multiplexing (OFDM) transmission.
 20. The method of claim 16, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.
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